Course project: Reconstruction of flow field for a lid driven cavity flow

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Project statement

The lid-driven cavity is a standard test case for verifying the accuracy of new computational methods for incompressible Navier–Stokes equations. The governing equation for steady state incompressible flow for a cavity reads

$$(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{Re}\nabla^2 \boldsymbol{u}, \qquad \text{in } \Omega \tag{1}$$

$$\nabla \cdot \boldsymbol{u} = 0, \qquad \text{in } \Omega, \tag{2}$$

$$\tag{3}$$

with prescribed boundary conditions as shown in Figure **??**.

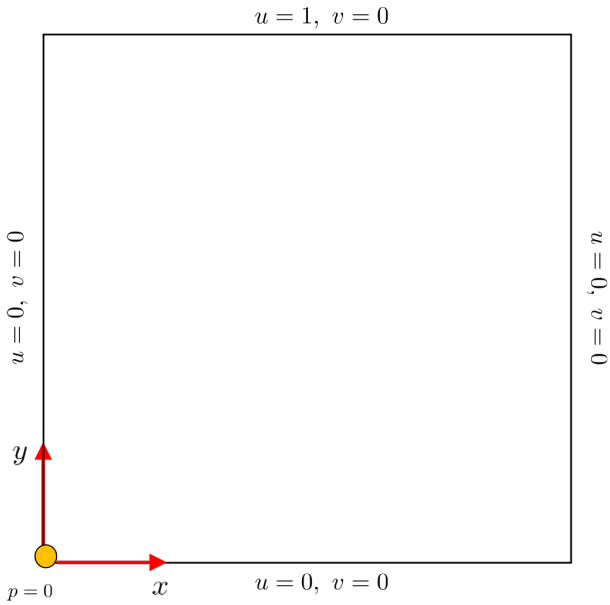

Figure 1: Domain and boundary condition for cavity flow

Figure **??** represents the domain with extent $[0, 1] \otimes [0, 1]$ and no-slip boundary conditions are applied at left, bottom and right boundaries. The top boundary is moving with constant velocity in the positive x direction. The solutions of $\boldsymbol{u} = [u, v]$ and p for $Re = 100$ is given in Figure 2.

Dataset

The data for $\boldsymbol{u}$ obtained by solving the **??** using spectral element method is provided. The data is provided in ".mat" format. The data can be loaded using *scipy* package of python, e.g. *scipy.io.loadmat*.

Important: You may use LLMs for *programming assistance only.* All physical reasoning, mathematical arguments, diagnostics, and interpretation must be your own. Grading emphasizes understanding of incompressibility, boundary layers, and reconstruction limits.

Task 0: Physics audit for cavity flow (required)

1. Describe the expected qualitative structure of the Re = 100 cavity flow: primary recirculation cell, secondary corner eddies, and shear layer near the moving lid. Indicate which features are driven

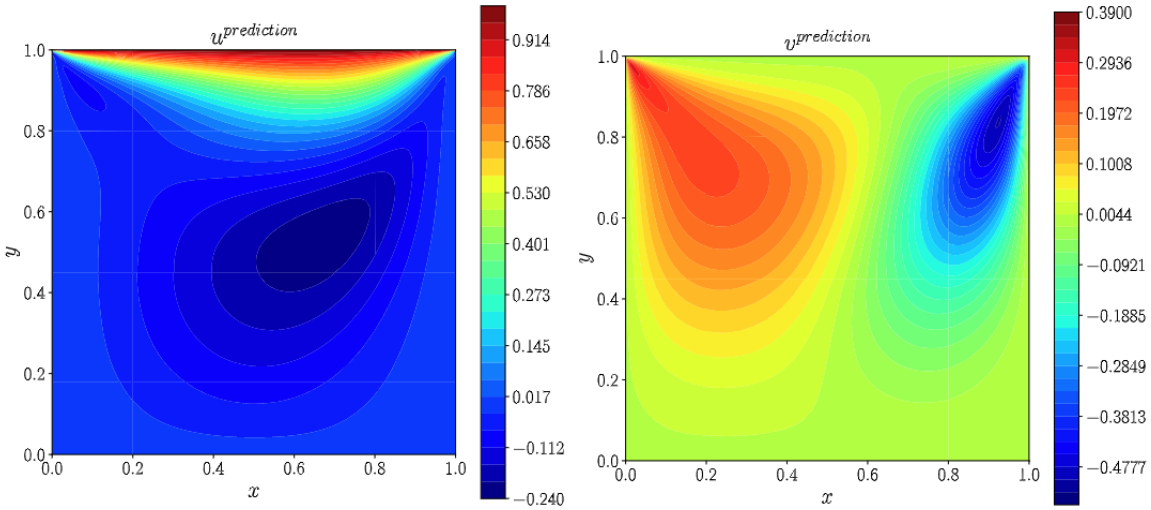

Figure 2: u, v, and p for a lid driven cavity flow

primarily by boundary conditions vs. nonlinear advection.

2. Explain why pressure is only determined up to an additive constant. Propose a gauge condition you will impose in your PINN implementation and justify it.

3. Write down at least two physically meaningful diagnostics beyond L^2 error that you will use in this project (e.g., divergence norm, vorticity extrema, centerline velocity profiles).

Task 1: Forward problem — solve with PINNs

1. Train a PINN to solve the steady cavity problem at Re = 100 using only the PDE residuals and boundary conditions.

2. Validate your solution by plotting:

   - $u(x, y)$ and $v(x, y)$ contour plots,
   - vorticity $\omega = \partial_x v - \partial_y u$,
   - centerline profiles $u(x, 0.5)$ and $v(0.5, y)$.

3. Compare your centerline profiles with those computed from the provided spectral-element dataset. Report relative L^2 errors for u and v.

4. **Failure-mode check (core):** Provide one example of a visually plausible PINN solution that is physically wrong (e.g., violates incompressibility locally or smears the lid shear layer). Give a diagnostic that detects it.

Task 2: Reconstruction from sparse interior data

1. Randomly sample $N = 10, 20, 30, 40$ points inside a fixed interior patch for measurements of u, v and p.

2. Reconstruct u, v, p over the entire domain using a PINN with data loss + PDE residual + boundary conditions.

3. Report relative L^2 errors for u, v, and p separately. Also report the divergence norm $\|\nabla \cdot \mathbf{u}\|_{L^2}$.

4. Plot error heatmaps and explain where (spatially) reconstruction fails first as N decreases (e.g., near corners, lid shear layer), and why.

Task 3: Sampling strategy and patch location

1. Compare two sampling strategies within the same patch: (i) uniform random sampling and (ii) Latin Hypercube Sampling (LHS).

2. Move the patch to two different locations: (A) near the center recirculation region, (B) near the lid shear layer.
3. For fixed N, compare reconstruction quality across these patch locations. Use your diagnostics from Task 0 to explain which region provides more information about the global flow.

Task 4: Noise robustness and regularization

1. Add synthetic noise (1%, 5%) to the interior measurements and repeat Task 2.
2. Propose and implement one robustness modification (e.g., Huber loss for data, adaptive residual weighting, smoothing/denoising of inputs, or collocation resampling).
3. Quantify how the modification changes reconstruction error and divergence error.

Task 5: Sensitivity to Reynolds number (conceptual)

1. Explain how the solution is expected to change qualitatively as Re increases from 100 to 1000 (e.g., thinner boundary layers, stronger corner eddies).
2. If you had sparse data at an unknown Reynolds number, what would be your strategy to infer Re? State what data (and where) would be most informative.

Submission requirements

- A report (8–10 pages) emphasizing: physics diagnostics, reconstruction limits, sampling/patch effects, and failure modes.
- Include contour plots, vorticity, centerline profiles, error heatmaps, and divergence diagnostics.
- Code may be assisted by LLMs; the written analysis must reflect your own understanding.

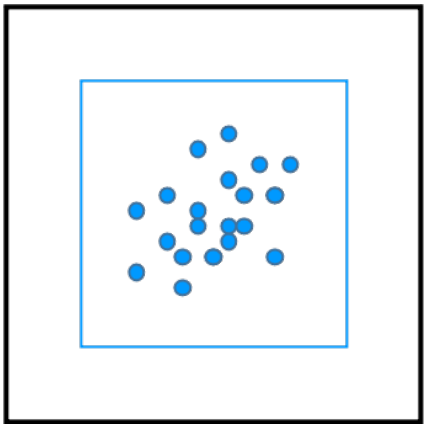

Figure 3: An example of patch and sample strategy.

Programming Options

You may use TensorFlow, PyTorch, DeepXDE or Modulus for completing the tasks.

References

[1] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.